ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

TECHNICAL PROGRAMMING

PAPER 1 · PRACTICE D

MARKS: 120 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Laid out like the DHET first paper (120 marks, Python). Section A (40): multiple choice with a pseudocode trace, true or false, choosing the correct term and matching · Section B (80): the CPU, binary and hexadecimal, waterfall and agile development, a while-loop trace, Git branches, an obstacle-distance logger with error handling to study, list and string coding, and inheritance with a Motor class.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. All program code is written in Python 3. Code is not marked strictly on syntax.
6. Write neatly and legibly.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Which part of the CPU carries out calculations and comparisons?

A Control unit
B ALU
C Cache
D Data bus

1.2 The decimal number 13 written in binary is …

A 1101
B 1011
C 1110
D 1001

1.3 Which Python function changes the text "25" into the whole number 25?

A str()
B float()
C len()
D int()

1.4 Which software development model delivers working software in short cycles called sprints?

A Waterfall
B V-model
C Agile
D Big bang

1.5 What is displayed by print(10 // 3)?

A 3.33
B 1
C 3
D 30

1.6 Which keyword is used to handle an error (exception) in Python?

A except
B elif
C pass
D global

1.7 Which method adds an item to the end of a list?

A insert()
B remove()
C pop()
D append()

1.8 How many different values can ONE hexadecimal digit have?

A 8
B 16
C 10
D 2

1.9 Which Git command shows the files that have changed but have not yet been committed?

A git clone
B git push
C git status
D git init

1.10 Study the pseudocode below. What is displayed?

```
x ← 2
FOR i ← 1 TO 4
    x ← x * 2
NEXT i
DISPLAY x
```

A 8
B 16
C 64
D 32

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.10) in the ANSWER BOOK.

2.1 The control unit of the CPU fetches and decodes instructions.

2.2 In Python, 7 % 2 gives 3.

2.3 A Python list can hold values of different data types.

2.4 The waterfall model makes it easy for the client to change the requirements at any time.

2.5 A syntax error stops a Python program from starting.

2.6 The hexadecimal digit A is equal to the decimal number 11.

2.7 A subclass can use the methods of its parent class.

2.8 The string method split() returns a list of strings.

2.9 The command git pull sends your commits to GitHub.

2.10 A Python function without a return statement returns None.

(10)

[10]

QUESTION 3

Choose the correct term from those in brackets and write it next to the question number (3.1–3.10) in the ANSWER BOOK.

3.1 The very small, very fast storage inside the CPU that holds the data it is working on is a (register/hard drive/flash drive).

3.2 The number system with base 2 is the (binary/hexadecimal/decimal) system.

3.3 A small piece of code written to check that one function works correctly is a (unit test/user story/storyboard).

3.4 The statement x = 5.0 stores a/an (float/integer/string).

3.5 The method that changes "robot" into "ROBOT" is (upper()/title()/capitalize()).

3.6 A loop inside another loop is a/an (nested/infinite) loop.

3.7 A group of values that must never change should be stored in a (tuple/dictionary/set).

3.8 Inside a method, self refers to the (object/module) itself.

3.9 The Git command that saves a snapshot of the staged changes is git (commit/push/clone).

3.10 In agile development, work is done in short cycles called (sprints/phases/releases).

(10)

[10]

QUESTION 4

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–O) next to the question number (4.1–4.10) in the ANSWER BOOK.

	COLUMN A		COLUMN B
4.1	Number system with base 16	A	constructor
4.2	Makes a copy of a GitHub repository on your own computer	B	len()
4.3	A class that inherits from another class	C	branch
4.4	Method that runs automatically when an object is created	D	hexadecimal
4.5	Breaks a string into a list of parts	E	logic error
4.6	A mistake that gives a wrong answer without stopping the program	F	syntax error
4.7	Block of code that runs only when an exception occurs	G	subclass
4.8	Returns the number of items in a list	H	split()
4.9	Document that lists everything the software must do	I	git clone
4.10	A separate line of development in a Git repository	J	except block
		K	requirements specification
		L	binary
		M	join()
		N	git push
		O	loop

(10)

[10]

TOTAL SECTION A: [40]

SECTION B

QUESTION 5

5.1 Name the THREE main parts of a CPU and give the function of EACH. (6)

5.2 Convert the following numbers. Show ALL working.

5.2.1 The decimal number 45 to binary (2)

5.2.2 The hexadecimal number 2F to decimal (2)

5.3 Explain the difference between the waterfall model and the agile model of software development. (4)

5.4 Study the Python code below and answer the questions that follow.

```
1  n = 5
2  result = 1
3  while n > 1:
4      result = result * n
```

```
5      n = n - 2
6  print(result)
```

5.4.1 What is displayed? (2)

5.4.2 How many times does the body of the loop run? (1)

5.5 Explain what a Git branch is and why programmers use branches. (2)

5.6 Explain the difference between git commit and git push. (2)

5.7 Give TWO advantages of storing a robot's program files in the cloud (for example on GitHub). (2)

[23]

QUESTION 6

A robot measures the distance to obstacles in front of it. Study the following program, which logs the distances, and answer the questions.

```
 1  readings = []
 2
 3  def add_reading():
 4      try:
 5          d = float(input("Distance in cm: "))
 6          if d < 0:
 7              print("Distance cannot be negative.")
 8          else:
 9              readings.append(d)
10      except ValueError:
11          print("Please enter a number.")
12
13  def report():
14      if len(readings) == 0:
15          print("No readings yet.")
16          return
17      print("Readings:", len(readings))
18      print("Closest:", min(readings))
19      print("Average:", round(sum(readings) / len(readings), 1))
20      close = [d for d in readings if d < 20]
21      print("Too close:", len(close))
22
23  while True:
24      choice = input("\nA Add  R Report  Q Quit: ").upper()
25      if choice == "A":
26          add_reading()
27      elif choice == "R":
28          report()
29      elif choice == "Q":
30          print("Goodbye")
31          break
```

6.1 What is the purpose of the try … except block in lines 4–11? (2)

6.2 What is displayed if the user enters ten as the distance? (1)

6.3 Explain what the return statement in line 16 does. (2)

6.4 Explain what line 20 does. (2)

6.5 The user adds the readings 35, 12.5, 18 and 54.5 and then chooses R. Write down the exact output of report(). (4)

6.6 Why is .upper() used in line 24? (1)

6.7 Write a function named clear_readings() that asks "Are you sure? (Y/N)" and empties the list only if the user answers Y. (5)

[17]

QUESTION 7

7.1 Write Python code to create a list named speeds with the values 120, 80, 150, 95 and 60 (motor speeds in r/min). (2)

7.2 Write code that displays the highest and the lowest speed, using max() and min(). (2)

7.3 Write code that sorts the list from the highest to the lowest speed and displays it. (2)

7.4 Write a function named average(values) that returns the average of the numbers in a list. (4)

7.5 Every robot has a serial code such as RB-2026-0045. Write code that asks the user for a serial code, uses split("-") to display the year part, and displays "Valid code" if the code starts with RB, or "Invalid code" otherwise. (6)

7.6 Write code that uses a for loop to count how many values in speeds are above 100, and then displays the count with a message. (6)

[22]

QUESTION 8

8.1 Define the term inheritance in object-oriented programming. (2)

8.2 Explain the difference between a method and an attribute, and give an example of EACH. (4)

8.3 Study the class below and answer the questions that follow.

```
class Motor:
    def __init__(self, name, max_rpm):
        self.name = name
        self.max_rpm = max_rpm
        self.rpm = 0

    def set_speed(self, rpm):
        if 0 <= rpm <= self.max_rpm:
            self.rpm = rpm
        else:
            print("Speed out of range")
```

8.3.1 Name the constructor of the class. (1)

8.3.2 Write down what is displayed when the following code runs: m = Motor("Left", 3000), then m.set_speed(3500), then print(m.rpm). (2)

8.4 Write the code for a subclass named ServoMotor that inherits from Motor. It must have an extra attribute angle (set to 0 in the constructor) and a method turn_to(angle) that changes angle only if the new angle is from 0 to 180, and otherwise displays "Angle out of range". (9)

[18]

TOTAL SECTION B: [80]

TOTAL: 120

END OF QUESTION PAPER